

(1) Five-card deck and bit encoding

Pick 5 cards: 3 hearts and 2 clubs. Encode a bit $b \in \{0, 1\}$ as a card pair:

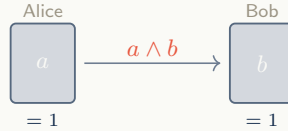
$0 \mapsto (\clubsuit, \heartsuit), \quad 1 \mapsto (\heartsuit, \clubsuit)$



Each card is a heart (1) or club (0); a bit is the order of its pair. The fixed middle heart makes three hearts, two clubs.

(2) Players' two secret bits a, b

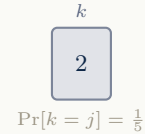
Alice commits a , Bob commits b . Alice's pair is negated so that the centre heart joins a triple iff $a \wedge b = 1$.



Row layout: $[\neg a_1, \neg a_2, \heartsuit, b_1, b_2]$ at positions $0, \dots, 4$.

(3) Pad: random cyclic shift k

Draw one pad value k uniformly from $\mathbb{Z}/5 = \{0, 1, 2, 3, 4\}$:

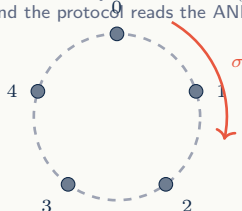


The pad is the entire randomness budget: one shift hides the row layout. Perfect security $\varepsilon = 0$ follows.

(4) Five cyclic positions, group C_5

The five card positions form one cycle under $\sigma = (0\ 1\ 2\ 3\ 4)$. The acting group is $C_5 = \langle \sigma \rangle$, and the protocol reads the AND output directly from this cycle.

Position action:
 $C_5 \hookrightarrow S_5$,
transitive,
regular.



Framework Klein bound (finite $\text{Aut}(\mathbb{P}^1)$ subgroups): $|C_5| = 5 \leq 60$, so the threshold check is trivial.

(5) Shuffle by σ^k , reveal, scan for $\heartsuit\heartsuit\heartsuit$

Rotate the row by σ^k and turn every card face-up. The AND output is the presence or absence of three consecutive hearts.

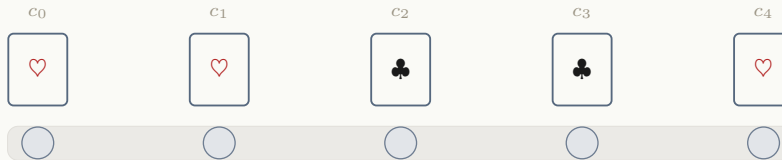


triple (wrap to c_4)

$a \wedge b = 1 \iff \exists i: c_i = c_{i+1} = c_{i+2} = \heartsuit \pmod{5}$

Security witness: the uniform cut certifies $\varepsilon = 0$ from the regular action of C_5 on the fibre.

(6) Reveal all 5 cards; every party reads $a \wedge b$



public reveal

All five c_j are public after the reveal. No private helper-shares: the protocol reconstructs by public broadcast, reading the bit straight off the cyclic row with no covering. $\varepsilon = 0$ holds because the uniform cut k fully randomises the card positions.